

Paramveer Singh

Machine Learning & Backend Engineer

Email: lco23586@ccet.ac.in — Phone: +91 97807 73834

[LinkedIn Profile](#)

[GitHub Profile](#)

[Portfolio](#)

[LeetCode](#)

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Summary

Machine Learning and Backend Engineer specializing in GPU-accelerated infrastructure, custom PyTorch/CUDA extensions, and stateful multi-agent RAG systems. Proven track record of building production-grade LLM inference architectures and highly scalable backends, driven by a passion for bridging the gap between frontier AI research and performant, secure engineering.

Experience

Bharat Electronics Limited

Jun 2026 – Present

Research And Development Intern

- Testing Digital Mobile Radio (DMR) systems including RF testing, ATP testing, and back-to-back testing.
- Performed physical and firmware debugging of critical communication hardware in high-reliability environments.

GirlScript Summer of Code

May 2026 – Present

Open Source Developer - AI Track

- Selected as an AI/ML Open-Source Contributor, optimizing machine learning models for community use.
- Conducted extensive code reviews and collaborated with a global team of developers to improve AI tooling.

CSIR – Central Scientific Instruments Organisation

May 2025 – Jul 2025

Software Development Intern

- Engineered a full-stack seat-allocation web portal (React.js, Next.js, MySQL) for the ISTC-CSIR government initiative, serving candidate and administrator portals.
- Implemented JWT-based authentication and automated seat-allocation algorithms, reducing manual processing time significantly for concurrent government users.
- Optimized backend database queries and redesigned real-time dashboard analytics, drastically improving page-load performance under heavy concurrent load.

Open Source Contributions

- **CERN (TIGRE & OpenPHIGS)** — *Links: [PR #748], [PR #749], [PR #31]*
Official Contributor (C/C++). Resolved severe stripe artifacts in parallel beam projections (Fixing intersection bounds) and corrected NumPy geometry instantiation bugs. Implemented robust GLU tessellation for concave polygon filling and integrated automated Doxygen API pipelines.
- **scikit-hep & CambridgeCIA** — *Links: [PR #1667], [PR #188]*
Engineered backward compatibility patches avoiding `std::any` in `uproot5` to support legacy ROOT versions for High-Energy Physics I/O. Added robust dynamic type and geometry introspection checks to the Learned Iterative Optimization Networks (LION) core model initializers.
- **FOSS United** — *Links: [PR #1631]*
Optimized the production dashboard by deduplicating redundant frontend API requests (Vue.js), heavily reducing server load and improving client-side state management.

Projects

FlashAttention-2 from Scratch (Flash-Atten2-MMA) — CUDA C++, PyTorch, Tensor Cores

Links: [GitHub]

- **Work:** Implemented FlashAttention-2 from scratch as a custom PyTorch C++ extension with dynamic kernel dispatching and Tensor Core MMA optimizations for D=128 LLM architectures.
- **Impact:** Achieved a **14.0x speedup** on BERT-large and **13.3x speedup** on LLaMA-7B prefill over the mathematically verified scalar loop baseline on NVIDIA T4.

Triton-Based Fused Operator Suite (Triton-MLOps) — OpenAI Triton, Python, vLLM

Links: [GitHub]

- **Work:** Engineered production-grade fused GPU kernels for LLM inference pipelines, aggressively fusing operations in SRAM to bypass PyTorch's ATen overhead.
- **Impact:** Fused Softmax reduced GMEM round-trips from 5 to 2. Prefix-Prefill Attention achieved **96.3% Time-To-First-Token (TTFT) reduction** at P=6144 on a Colab T4 GPU.

Custom HGEMM (Half-Precision GEMM) — *CUDA C++, CuTe, WMMA*

Links: [\[GitHub\]](#)

- **Work:** Built a 7-phase iterative HGEMM kernel using SMEM tiling, Tensor Cores, async pipelining, and XOR-swizzling to eliminate shared-memory bank conflicts.
- **Impact:** Approached theoretical peak performance, achieving **13.3 TFLOPS (70.2% cuBLAS efficiency)** on NVIDIA T4, scaling from 1.9% in the naive baseline.

GateKeeper (LLM API Gateway) — *Go, Redis, NATS JetStream, PostgreSQL*

Links: [\[GitHub\]](#)

- **Work:** Built an API Gateway and intelligent rate limiter purpose-built for LLM traffic with token-aware limiting, failover routing, and asynchronous cost tracking.
- **Impact:** Processed **┆5,000 RPS** with **┆10ms p50 latency overhead**. Implemented Semantic Vector Caching (Redis FT.SEARCH) yielding **┆5ms responses** and 100% token cost savings for repeated queries.

microgpt — *Python, PyTorch*

Links: [\[GitHub\]](#)

- **Work:** Developed a compact autoregressive language model from first principles incorporating modern 2025-era architectures (GQA, RoPE, SwiGLU, RMSNorm).
- **Impact:** Engineered for scalable distributed training, eliminating 2017 Transformer inefficiencies to match Llama 3's high-performance architectural design constraints.

Semantic-search-RAG — *Python, FastAPI, Kafka, PySpark, Elasticsearch*

Links: [\[GitHub\]](#)

- **Work:** Engineered a distributed microservices platform for Enterprise Semantic Search, featuring Hybrid Search (BM25 + Dense Vectors + custom SPLADE v2).
- **Impact:** Ingested millions of documents through a robust streaming pipeline, delivering **sub-100ms** LLM-powered answer generation decoupled via gRPC embedding services.

Publications

- **IEEE-ICSPED 2025:** *CHRONOS: A Chaos-Harnessed TRNG for Neural Initialization and Secure Key Synthesis*. Designed a True Random Number Generator passing all 15 NIST SP 800-22 tests. [\[Publication Link\]](#)
- **Springer Journal 2025:** *AI-Driven Sanskrit Character Recognition and Translation...* End-to-end ML pipeline with Tesseract OCR, custom HuggingFace seq2seq, and Vedic TTS. [\[Publication Link\]](#)

Scholastic Achievements

- **NASA Space Apps Challenge 2025:** Global Nominee & Regional Winner (Selected from 18,868+ teams).
- **IPD EXPO 2025:** 1st Place Overall (Best technical execution across 30+ projects).
- **India Innovates 2026:** National Finalist (Recognized among 35,000+ national entries).
- **AI Hack Matrix 2026:** Finalist, Chandigarh University (Advanced from 300+ teams).

Skills

Languages: Python, C/C++, CUDA C++, Go, JavaScript, SQL, Shell

ML & GPU: PyTorch, OpenAI Triton, HuggingFace, LangGraph, OpenCV, Tensor Cores (wmma)

Backend & Cloud: FastAPI, React.js, Next.js, Docker, Apache Airflow, AWS, GCP, Git

Education

Chandigarh College of Engineering & Technology, Panjab University

Aug 2024 – 2027 (Exp.)

B.E Electronics and Communication Engineering — Score: 7.4/10

CSIR-CSIO Indo Swiss Training Centre

Jul 2020 – Jul 2023

Diploma of Mechanical Engineering (Tool and Die) — Grade: 8.5 CGPA

Certifications

- **Quantum Cryptography** (QNu Labs)
- **Model Context Protocol (MCP)** (Hugging Face)
- **Linux Kernel Development (LFD103)** (Linux Foundation)
- **Machine Learning Onramp** (MathWorks)